Prime-field k-sum tables against a tuned mitm incumbent, on 16 fresh curves

2026-10-30

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Track: generic decomposition (bounded by √r; cannot beat rho). Both arm families are generic decomposition oracles (AGENTS.md "Tracks of index calculus", 2026-10-29). Nothing here moves the algebraic track.

Status: UNQUALIFIED. One host, one binary, no `claims/*.json` record. The second-host replay of #2866 is in flight (`research/prime_ksum_vs_incumbent_20261029/second-host/`); this study's own holdout has not been replayed on another host.

Result. The ranking of #2866 survives fresh curves, pooling and tuning. On 16 fresh curves (4 per size, 20 to 32 bits), with the incumbent re-tuned on separate train curves by the same rule as the k-sum tables, the k = 5 sum table costs 0.70×, 0.45×, 0.32× and 0.22× the tuned incumbent at 20, 24, 28 and 32 bits. The registered primary test holds. The table is paired ratios of total cost at w = 0, with the cluster-bootstrap 95% interval over targets in brackets.

bits	`ksum5-sel / inc-sel`	`ksum4-sel / inc-sel`	registered range (k = 5)
20	0.700 [0.682, 0.718]	0.632 [0.611, 0.654]	0.65–0.95
24	0.446 [0.432, 0.461]	0.452 [0.440, 0.465]	0.40–0.55
28	0.324 [0.318, 0.331]	0.336 [0.329, 0.344]	0.30–0.45
32	0.217 [0.212, 0.222]	0.253 [0.248, 0.258]	0.20–0.32

- **Every curve, every weight.** At each size the k-sum arms are below the tuned incumbent on **all 4 curves** (16 of 16), and at all three band weights in the pooled test.
- **Against the incumbent's best at w = 0.** At 24, 28 and 32 bits the k = 5 ratio to `inc-w0` (the incumbent cell with the lowest train S at w = 0) is 0.486, 0.368 and 0.248.
- **Exponent (toy tier, w = 0).** `ksum5-sel` 0.496 [0.487, 0.510], `ksum4-sel` 0.525 [0.516, 0.537], tuned incumbent 0.635 [0.627, 0.645]. For reference, rho-neg is 0.457 and bsgs-neg 0.501 on the same sessions.

Distance remaining (pooled, w = 0): `ksum5-sel` is 6.0×, 5.8×, 7.3× and 7.3× rho-neg, and 7.6×, 6.5×, 7.2× and 7.1× bsgs-neg at 20, 24, 28 and 32 bits. The method is generic, so its limit is BSGS, and it stays above BSGS at every size.

Public synthetic curves and targets; no security claim. Registered in `PROTOCOL.md` before any timed run (`REGISTRATION.sha256`); the selection and the holdout specs were hashed before the holdout ran (`select/FROZEN.sha256`).

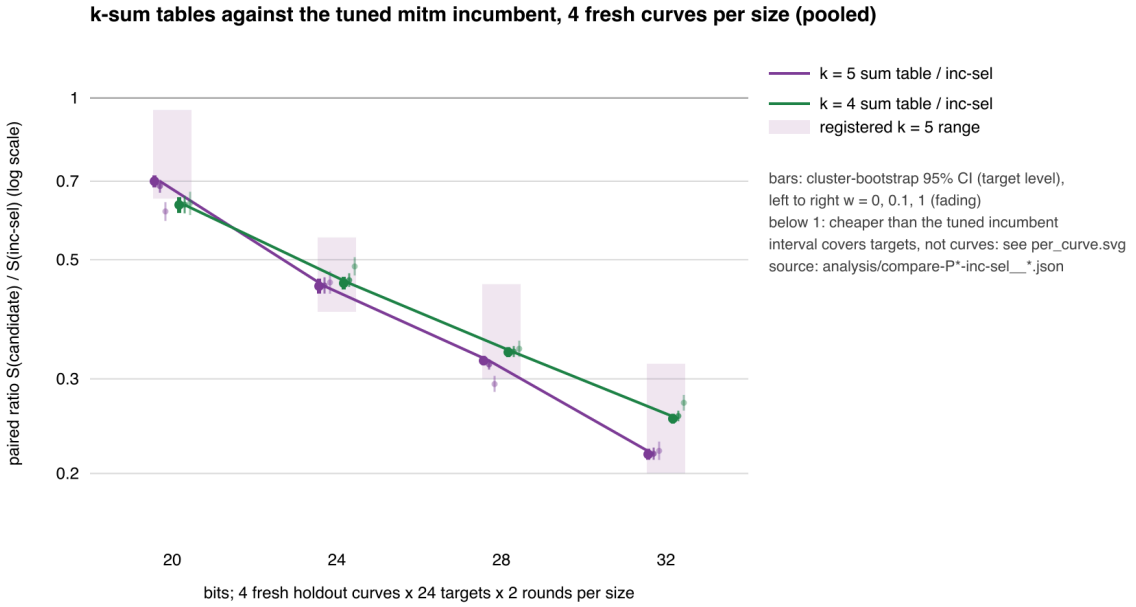


Figure 1: Pooled paired ratio of each k-sum arm to the tuned incumbent per size, at w = 0, 0.1 and 1, with the registered k = 5 range. Source: ratio.svg.

1. What was asked, and the answers `[measured]`

Follow-ups 2 to 4 of the #2866 list.

question	answer
Q2. Does the ranking hold on fresh curves, pooled?	**Yes.** 16 of 16 fresh curves, and the pooled primary test at 24, 28 and 32 bits (and at 20).
Q3. How much of the gap is tuning?	**Very little at w = 1.** The incumbent's selection at w = 1 is the recorded base at 24, 28 and 32 bits, so `inc-sel` is the recorded recipe there. Tuned for w = 0 the incumbent is 8–13% cheaper (derived below), and k = 5 is still 0.25–0.49× of it. At 20 bits the grid's edge was picked (limits).
Q4. Does a train-frozen rule choose k?	**Yes at w = 1, with a caveat at 20 bits.** The rule chose k = 5 at every size. On the holdout k = 5 is not above k = 4 at w = 1 (at 24, 28 and 32 bits it is below). At 20 bits k = 4 is cheaper at w = 0 and 0.1, and the two are unresolved at w = 1.

2. Method `[measured]`

- **Curves.** All fresh `prime_search` curves. The 16 holdout curves share nothing with the 24 earlier curves compared (`plans/freshness.txt`): this study's train curves, the four curves of #2866, the #2852 train and test panels, and the incumbent's train sessions.
- **Train.** 2 curves, 12 targets, 2 rounds per size. Each family got 5 sizes of base, 0.5×–2× its anchor. Per family, the cell with the lowest mean S at **w = 1** was selected. The same rule applies to the incumbent and both k-sum families; the ties rule and the grids are in `PROTOCOL.md`. The selection is in `select/selection.json`.

bits	`inc-sel` N	`inc-w0` N	`ksum4-sel` A	`ksum5-sel` A	k*
20	64 (grid edge)	64	10	9	5
24	162	115	14	11	5
28	407	289	21	15	5
32	1024	727	30	19	5

- **Holdout.** 4 curves, 24 targets, 2 rounds, 1 warm-up per size. Arms: `rho-neg`, `rho-neg-aa`, `bsgs-neg`, `inc-sel`, `inc-w0` (when its N differs), `inc-rec` (the recorded recipe of #2866), `ksum4-sel` and `ksum5-sel`. Every run is one cold target, with the table built and charged inside the run.

- **Binary.** `ecbench-v2` (`BINARY.sha256`), the binary of #2852 and #2866.

3. Integrity `[measured]`

- **Every answer verified, every replay reproduced.** The train sessions have 4,896 records and the holdout sessions 8,928, all verified. `ecbench verify --replay-all` reproduces every replay with 0 problems (`audits/`).
- **A/A.** `rho-neg` vs `rho-neg-aa` is unresolved inside the band at every size.
- **Interval scope.** `ecbench compare` reports its interval over workloads (targets), also on these pooled sessions (`analysis/interval-scope.txt`). So the pooled interval does not carry between-curve variance. As registered, robustness across curves is judged by the per-curve test below, and the primary verdict needs it.

4. Registered tests `[measured]`

Primary: holds at 24, 28 and 32 bits. `ksum5-sel` and `ksum4-sel` are below `inc-sel` at all three weights, with the interval's upper bound below 1 at w = 0, **and** below `inc-w0` at w = 0, 0.1 and 1.

Secondary.

- **(a) Per-curve consistency, holds at every size.** The k-sum family with the lower pooled S is below `inc-sel` on 4 of 4 curves (and below `inc-w0` on 4 of 4 at 24–32 bits). Per-curve table: `analysis/per_curve.md`.
- **(b) 20 bits, holds:** 0.700 for k = 5 and 0.632 for k = 4 at w = 0, below 1 at every weight.
- **(c) Recorded incumbent and tuning.**
  - The incumbent's w = 1 selection equals the recorded base at 24, 28 and 32 bits, so `inc-sel / inc-rec` is exactly 1.
  - At 20 bits, N = 64 against the recorded 45 is unresolved: 1.034 [1.002, 1.066] at w = 0, 1.000 at w = 0.1 and 0.863 [0.830, 0.898] at w = 1.
  - The w = 0-tuned incumbent is cheaper at w = 0 than the recorded recipe. Derived from the ratios of the two k-sum arms to each of them: 0.92, 0.88 and 0.88 at 24, 28 and 32 bits, outside the 0.9–1.05 range registered for the w = 1 selection. This is not a direct comparison.
- **(d) Rule for k: holds at every size**, scored at w = 1 on the pooled interval of `ksum5-sel / ksum4-sel`:

bits	k*	`ksum5-sel / ksum4-sel`, w = 1	reading
20	5	0.965 [0.918, 1.013]	unresolved
24	5	0.934 [0.893, 0.975]	k = 5 below
28	5	0.860 [0.827, 0.893]	k = 5 below
32	5	0.814 [0.787, 0.843]	k = 5 below

At 20 bits the w = 0 ratio is 1.106 [1.070, 1.142], so k = 4 is cheaper at the lightest pricing. The registered w = 1 rule is the conservative one, and the 20-bit choice depends on the weight.

- **(e) Exponent.** Toy tier, 16 curves × 4 sizes, every fit admissible:

arm	w = 0		w = 0.1		w = 1	
`inc-sel`	0.635	[0.627, 0.645]	0.634	[0.628, 0.644]	0.631	[0.626, 0.638]
`ksum4-sel`	0.525	[0.516, 0.537]	0.526	[0.517, 0.535]	0.527	[0.518, 0.535]
`ksum5-sel`	0.496	[0.487, 0.510]	0.497	[0.489, 0.509]	0.504	[0.496, 0.510]

`ksum5-sel` is below `inc-sel` by 0.139 at w = 0, past the 0.1 registered. The fit is over four sizes (20 to 32 bits), so it is a small-size exponent, not the asymptotic 5/9 of the generic optimum. Full table: `analysis/exponents.md`.

- **(f) Distance remaining.** See the results paragraph above.

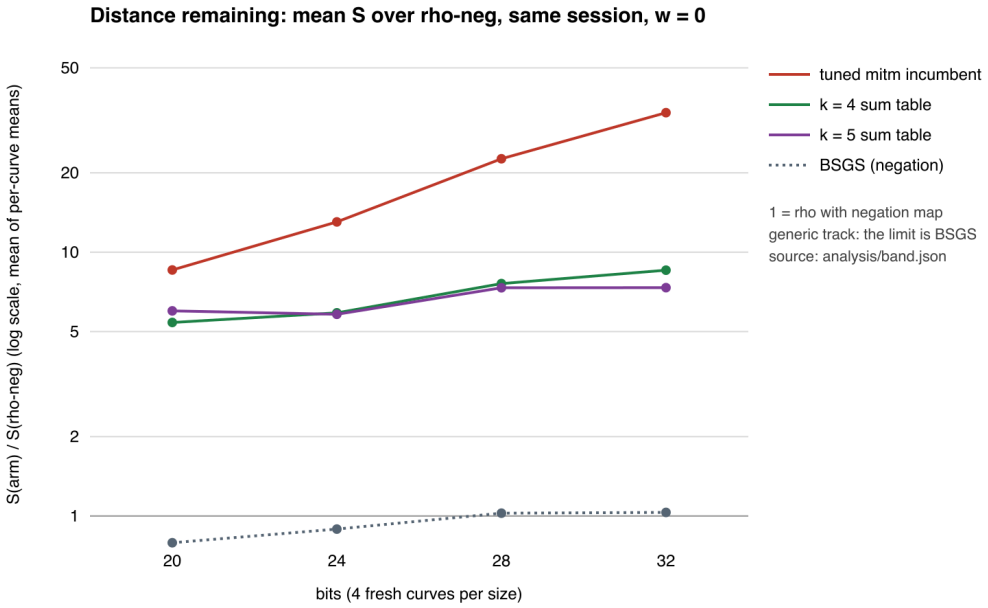


Figure 2: Mean S over rho-neg for each arm, same session, w = 0. Source: rho_multiple.svg.

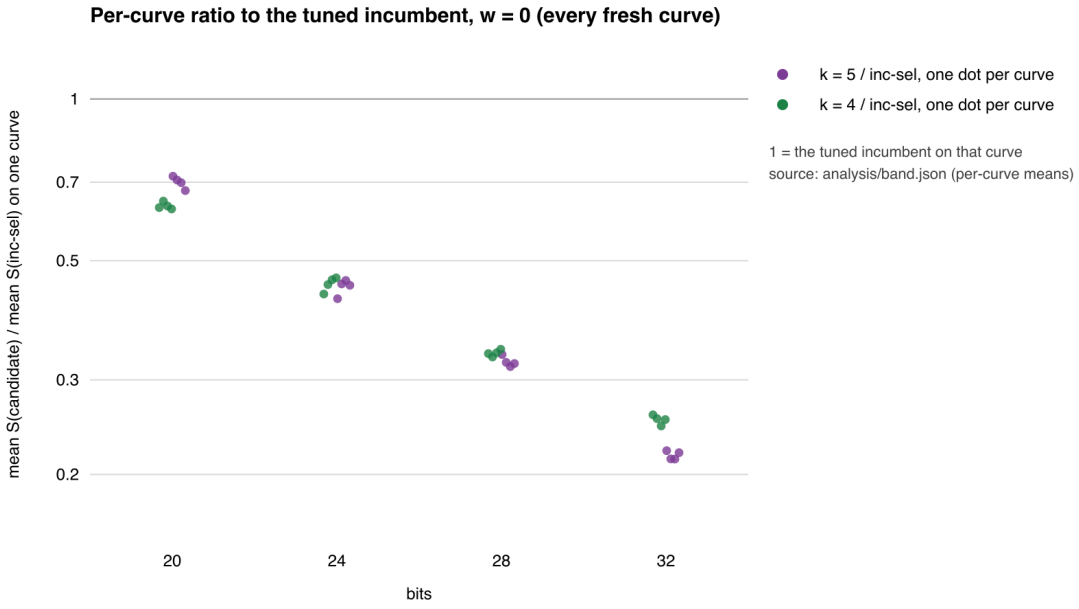


Figure 3: The per-curve ratio of each k-sum arm to the tuned incumbent, w = 0, on every holdout curve. Source: per_curve.svg.

5. Predictions scored

registered prediction	outcome
`ksum5-sel / inc-sel` in 0.65–0.95 / 0.40–0.55 / 0.30–0.45 / 0.20–0.32	all four inside
`inc-sel / inc-rec` in 0.9–1.05 at 24, 28, 32 bits	1.000 (the selection kept the recorded base)
k = 4 chosen at least once at 20 or 24 bits; k = 5 at 28 and 32	**missed:** k = 5 was chosen at every size
`ksum5-sel` exponent below `inc-sel` by at least 0.1	0.139
k-sum stays above BSGS at every size	yes, 6.5–7.6×

6. Reading

1. **Improvement of the generic-track baseline, on fresh curves, against a tuned incumbent.** The k = 5 table costs 0.70×, 0.45×, 0.32× and 0.22× the tuned incumbent at 20 to 32 bits, on every one of 16 curves. Class: engineering within the generic track.
2. **The #2866 gap is the method.** Tuning the incumbent at w = 1 changes nothing at 24 to 32 bits, and tuning at w = 0 moves it by 8–13%, against a 2.2–4.6× gap. The widening gap shows up in the exponent: 0.496 against 0.635.
3. **The 20-bit cell is the fragile one.** The incumbent's pick sat on the edge of the grid, its tuning there is unresolved, and the choice of k depends on the weight. Open #2826 (stride-renew, a reported 16% cut at 20 bits) would move 0.70 to about 0.84 if it merges. Still below 1.
4. **Neither the algebraic track nor rho is touched.** The best arm is still 5.8–7.3× rho on these curves.

7. Limits

- **Grid edge.** The 20-bit incumbent pick (N = 64) is on the edge of the grid, and no extension was run, as registered. The incumbent could be cheaper at larger N there.
- **One host.** UNQUALIFIED until a second-host replay and a `claims/*.json` record. The replay of #2866′s sessions is in flight; this holdout has not been replayed elsewhere.
- **Counted unit only.** ecbench GAE with §5a pricing of native work; the table lookups are hash probes priced in that band. Wall time is recorded in the sessions and not analysed.
- **Four sizes, 20 to 32 bits.** The exponent is a small-size fit. No claim above 32 bits.

8. Files

- `PROTOCOL.md`, `REGISTRATION.sha256`, `registered-at.txt`, `BINARY.sha256`: the registration. `make_specs.mjs`, `select.mjs`: tooling hashed in it.

- `specs/train-P<b>.json`, `specs/holdout-P<b>.json`, `plans/`, `select/`: the panels, resolved plans, the frozen selection and its hashes.
- `run-train.sh`, `run-holdout.sh`, `analyse.sh`: thin `ecbench` orchestration. `ECBENCH=` selects the binary.
- `sessions/`: native session records. The repository bot moves records over 64 KiB to the artifact store.
- `audits/`: the `verify --replay-all` receipts.
- `analysis/`: `band.json/md`, the paired comparisons, `ratios.md`, `per_curve.md`, `verdicts.json`, `exponents.md`, `bounds/`.
- `tables.mjs`, `plot.mjs`, `ratio.svg/png`, `rho_multiple.svg/png`, `per_curve.svg/png`: tables and figures (tooling).
- `dashboard.mjs`: the §7a update (tooling, no Python). One progress point for the best cold one-target IC cell of the holdout panel, whatever the outcome.
- `second-host/`: the replay specs and runbook of #2866′s study; see its README for the launch state.
- `report.pdf`: this README.